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A numerical investigation into the influence of soil type and root architecture on tree anchorage

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Abstract

The influence of root morphology and soil type on the mechanical behaviour of tree anchorage was investigated through numerical modelling. We developed a simple computer program to construct three-dimensional virtual root architectural patterns. This tool was used to build four schematic patterns: heart-, tap-, herringbone- and plate-like root systems. Each of these rooting types was characterized by specific branching characteristics. However, the total volume (proportional to the wood biomass) and material properties were kept constant. The finite element method was used to calculate the mechanical response of root/soil systems when the stem was subjected to bending forces. The overturning resistance of the four schematic root patterns was determined in four different idealistic soil types. These soils were based on Mohr–Coulomb plasticity models. Results showed that soil internal friction modified the position of the rotation axis during tilting of the root/soil plate. Rooting depth was a determinant parameter in sandy-like soils. Overturning resistance was greatest in heart- and tap-root systems whatever the soil type. However, the heart root system was more resistant on clay-like soil whereas the tap root system was more resistant on sandy-like soil. Herringbone and plate root systems were twice as less resistant on clay soils and 1.5 times less resistant on sandy soils when compared to heart and tap-like structures.

Introduction

The recent wind storms experienced throughout Europe have caused much damage and huge economic losses in both plantation and natural forests (Bergonzini and Laroussinie, 2000; Bouchon, 1987). If predictions that our climate is becoming windier prove to be correct, such disasters are likely to become more frequent in the future. When trees are subjected to strong winds, various types of damage are observed, however, failure usually occurs through uprooting (Cucchi and Bert, 2003; Nicoll et al., 1995).

Traditionally, vulnerability to uprooting has been investigated at the population level, by correlating forest damage to certain stand characteristics i.e. tree species, sylviculture treatment and practice, soil type and wind speed (Putz et al., 1983; Ruel et al., 2000). However, such an empirical analysis of windthrow is incomplete due to the numerous variables influencing tree stability, which are not often taken into account, e.g. root system or soil characteristics and soil hydrology.

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Experimental studies can provide a certain amount of detailed data concerning tree mechanical stability. Typical tests include winching trees sideways until failure and measuring the force necessary to cause failure (Cucchi et al., 2004; Peltola et al., 2000; Stokes, 1999). However, even if root system architecture is measured, data are often incomplete as roots are usually damaged during the tests (Stokes et al., 2006, b).

In order to improve tree stability in a plantation, an understanding of the physical processes occurring in the soil is also needed. The root/soil plate of trees is a compound structure, therefore both the physical properties of soils and structural properties of the tree, e.g. root architecture are important in determining the overall anchorage capacity of a tree. The type of failure during a storm depends not only on soil hydrology, but also on soil type. Research has focused on the investigation of fundamental mechanisms involved during uprooting by using mechanical models (Blackwell et al., 1990; Dupuy et al., 2005a, b; England et al., 2000; Ennos, 1990; Mattheck and Breloer, 1994). However, no realistic model yet exists which considers both the soil and the complete root system architecture as distinct elements.

The difficulty of investigating root anchorage is not only due to the complexity of the mechanisms occurring in both roots and soil, but also to their multifactorial aspect (Stokes et al., 2006). A good alternative to difficult and time-consuming field experiments is numerical modelling which will also help with the interpretation of experimental data or aid in identifying the parameters needing greatest study. Dupuy et al. (2003) carried out such analyses using the Finite Element Method (FEM), a powerful numerical method used in engineering. These authors developed methods allowing morphological data from real or simulated root systems to be subjected to virtual uprooting tests. Soil mechanical properties could also be changed easily, thereby allowing a rapid assessment of root anchorage in different soils. In a similar numerical investigation, Dupuy et al. (2005a) performed sensitivity analyses on the geometrical, topological and constitutive parameters which affect uprooting. These authors showed that the most important factors governing individual root resistance to pull-out in a given soil, were a combination of root branching parameters and morphology. However, this study was carried out using two-dimensional (2D) models at a very local level, i.e. on small root components approximately 0.1-m long with few branches.

Investigating the anchorage of adult trees is rather more complicated due to the morphological complexity encountered in large tree root systems (Danjon et al., 2005). Several structural elements of importance for mechanical stability can co-exist. Coutts (1983, 1986) identified the main components that play a role in the root anchorage of shallowly rooted Sitka spruce (*Picea sitchensis* Bong. Carr). Coutts (1983, 1986) showed that the windward lateral roots held in tension were the most important component of root anchorage, followed by the weight of the root/soil plate, soil cohesion and the bending strength of leeward roots. However, no tap roots were present in these trees. Tap roots are the large, central roots found in root systems of certain species e.g. *Pinus pinaster* Ait or in young trees (Hintikka, 1972; Köstler et al., 1968; Danjon et al., 2005). In winching experiments on different tree species, Stokes et al. (2000; 2005, 2006) found that the most resistant species to overturning were those possessing a 'heart' root system, i.e. with many branched, oblique, horizontal and lateral roots (Köstler et al., 1968). Tap-rooted systems were also tested and were more resistant to overturning when compared with shallowly rooted plate type systems e.g. Norway spruce (*Picea abies* L.). However, root architecture was not accurately measured in these species, therefore, the exact role of either morphology or topology i.e. the way branches are linked together, is not known.

One further parameter which is rarely measured and understood even less is the relationship between root architecture and potential anchorage, particularly with regards to different soil types. As experimental studies are difficult to carry out and interpret, modelling root architecture and numerical analyses may help to fill this gap in the knowledge (Fourcaud et al., 2003a). It is currently not known whether root architecture or soil type influences uprooting most, and it is likely that each parameter is dependent on the other.

The aim of our study was to determine the anchorage efficiency of four typical root system architectures in theoretical soils representing a

wide range of mechanical characteristics. For this purpose, a program was developed to construct simple three-dimensional architectural patterns. The four different root system types were then generated using real data (Dupuy, 2003) as input to the program, considering a small number of parameters. More realistic structures could also be implemented using existing models (Jourdan and Rey, 1997; Pages et al., 2004). Nevertheless, the complexity of these models would not enable us to easily impose given morphological characteristics of root components. Three of the patterns chosen were typical of those found in different forest tree species, and the fourth type, 'herringbone' (Fitter, 1990 ; Stokes et al., 1996) was an extreme example used to test the model, but which occurs in young plants and mature trees grown from cuttings e.g. poplar (*Populus* sp.) (Dupuy, 2003). FEM was then used to simulate the mechanism of uprooting for these different root types in idealistic sandy- and clay-like soils of different mechanical properties. The results should enable foresters and tree breeders determine the most important characters of tree stability with regards to root architecture and will also indicate how to manipulate these characteristics depending on soil conditions.

Materials and methods

Generation of schematic root systems

The variability of morphological patterns within root systems, e.g. spatial distribution of branching, branching angle, root length and diameter, needs to be considered carefully when analyzing the effect of root architecture on the uprooting process. The use of models is highly beneficial, as they enable deterministic structures to be created, for which the morphological parameters can be controlled. Therefore, we developed a program called SIMUL3R which generates typical root structures using data from real root systems as input. It must be underlined that this program does not simulate dynamic root growth processes, in the sense that chronological events are not taken into consideration, but it can be used to build root systems at a given stage of their development.

Description of the SIMUL3R program

The SIMUL3R program is a stepwise builder that reproduces – in a certain way – the elementary processes of primary elongation and ramification. However, the program does not consider either root mortality or the growth of adventitious roots. The root structure was described as a set of single roots connected together at particular insertion points. The growth of each individual root resulted from the activity at its apex. Root axes were split into unit segments of length Δl and diameter d. The points defined by the segment extremities were called nodes. The distance between two successive nodes bearing lateral ramifications was denoted ΔL (Figure 1).

The program which generates root architecture was based on three main modules (Figure 2): *INITIALISATION*, *EXTENSION*, and *VOLUME*.

The *INITIALISATION* stage consisted of defining a preliminary pool of N_0 apexes, i.e. potential roots, which would develop from the stump positioned at the origin (0,0,0). This set of initial apices contained a fixed number of laterals and an optional tap root.

The second module *EXTENSION*, was based on a stepwise procedure where the activity of each apex was computed independently from the others. At each step, the apex activity corresponded to the creation of a new root segment and any new lateral axes, i.e. new apices. The diameter of each segment was defined using the taper properties of the mother root. These root characteristics were determined at each step using morphological functions: *f_growthI(X)*, *f_ram(X)*, *f_ratio(X)* and *f_taper(X)*. The input parameters of these functions could be related to any variable X, e.g. time, branching order or spatial position. These functions can be either deterministic or stochastic and could be fitted statistically onto real root systems (Dupuy 2003), or they could be purely theoretical.

At each calculation step n, the current apex created a new root segment. Running of the primary growth function *f_growthI(X)* allowed the root increment Δl_n^m to be determined in length and direction. This apex could also initiate new branches during its extension according to the ramification function *f_ram(X)* that calculated the distance ΔL_d^m between two successive branching nodes, as well as the direction of the

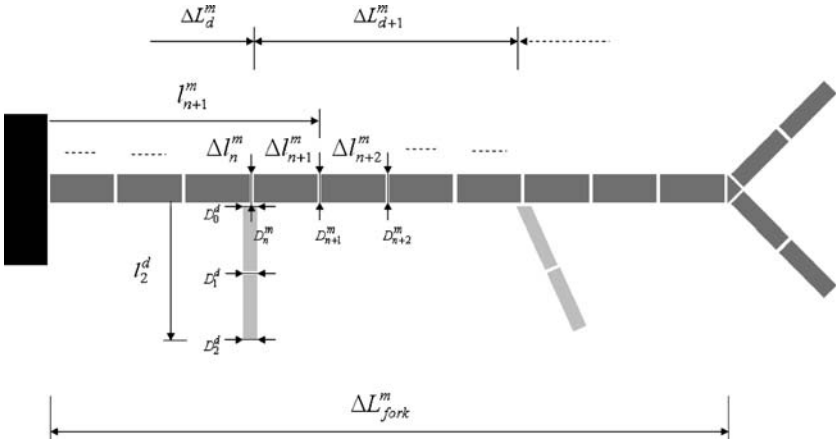


Figure 1. Models of root architecture were used to build typical three-dimensional root structures. Simulated root systems, were described as a set of connected unit segments of length Δl. The nth segment (created at step n of the simulation) of the mth root Δl_n^m is characterized by initial diameter D_n^m and final diameter D_{n+1}^m. The length of root at step n is written l_n^m. The mth root may initiate lateral branches i.e. dth root on the figure. The distance between two consecutive branches on the nth root is denoted ΔL_I^m. The mth root may also end with a fork. ΔL_{fork}^m denotes the distance of the fork from the insertion point.

new daughter lateral segment Δl_0^d. Distal roots could also be forked at their extremities. An appearance of such a fork was defined by its distance ΔL_m^{fork} to the root insertion node. The procedure that generated forks was similar to those building lateral branches, except that it caused the death of the bearing root apex. Root diameters were not calculated in the *EXTENSION* procedure. Nevertheless, the *f_ratio(X)* function was used to determine the theoretical coefficient giving the ratio between the initial diameter D_0^d of a root d and the diameter D_n^m of the mother bearing segment m (Figure 1). Similarly, the function *f_taper(X)* furnished the taper coefficient linking the diameter of successive segments along the root. These ratio coefficients were used by the *VOLUME* module in order to calculate the diameters of all segments at the final stage of the simulation. The building of a single root was stopped when predefined spatial boundaries were reached. The *EXTENSION* module ended when the set of apices to be treated was empty.

At this stage, only the root system skeleton was available. The *VOLUME* module was finally called in order to calculate root diameters when a predefined volume of the whole root system was expected. The diameter of all the root segments was linked together according to the ratio coefficients previously determined during the *EXTENSION* procedure. For that reason, the only variable to be calculated to fit the whole root volume was the bole diameter D_0^0. This initial diameter was determined using a dichotomy algorithm (a heuristic method of successive calculations) so as to obtain the expected root system volume. Finally, the root structure was recorded under a specific file format allowing three-dimensional visualization, architectural analysis and mechanical calculations to be performed.

Computation of schematic root patterns

Various types of root structures can be constructed using the SIMUL3R program, by varying the morphological functions *f_growth*I*(X)*, *f_ram(X)*, *f_ratio(X)* and *f_taper(X)*. In our study, simplified structures were used to represent different types of rooting patterns. Therefore, by using such simple representations, we can determine the contribution of certain root morphological characteristics to anchorage efficiency (Ennos, 1990; Niklas et al., 2002). Table 1 shows the characteristics of the morphological functions that were used to generate the schematic heart-, tap-, herringbone- and plate-like root systems. These data were taken from measurements of real root systems of adult trees of different species described elsewhere (Dupuy, 2003). The heart root system was typical of that found in e.g. Red oak (*Quercus rubra* L., Lyford, 1980) with dense branching and a large quantity of forks present; the tap root system

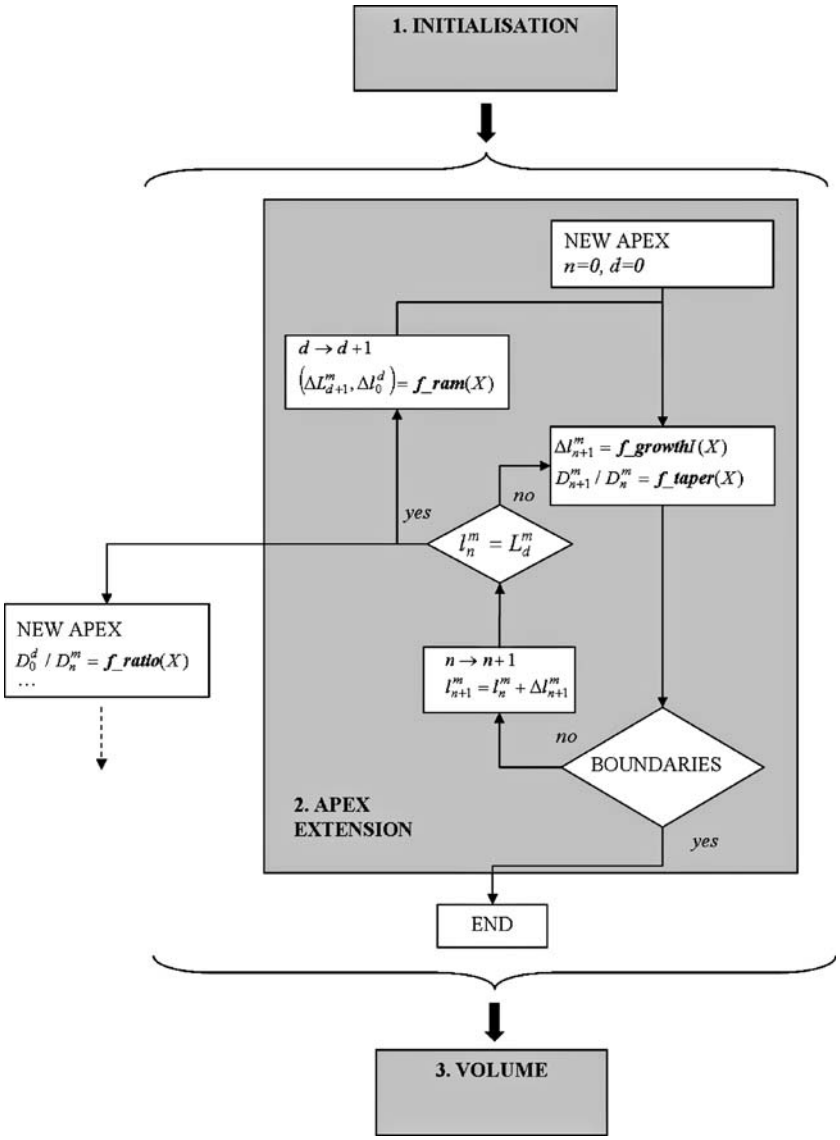


Figure 2. Flowchart of SIMUL3R program for the simulation of root architecture. The program is initiated with the *INITIATION* module which defines a set of apices. The *EXTENSION* module is then used to generate root apex activity in terms of extension and branching. At each step of the procedure, each apex operates an extension of length Δl in the direction given by the model *f_growth*I. When branching occurs $l_n^m = L_I^m$, a new apex is created and its initial diameter D_0^d is calculated from the *f_ratio* function. The function *f_ram* calculates the distance ΔL_{i+1}^m at which the next branch will be inserted and its direction. When predefined boundaries are reached e.g. maximum depth, radial distance or branching order, apex activity stops. After all apex activity has stopped, the *VOLUME* module enables the calculation of diameters in the root system in order to get a fixed total volume of root.

was generated using data from Maritime pine (*Pinus pinaster* Ait., Danjon, 1999); the herringbone system can be found in mature poplar (*Populus* sp.) trees originating from cuttings (Dupuy, 2003) and the plate-like system was typical of that found in e.g. Norway spruce (*Picea abies* L., Puhe, 2003, Stokes et al., 2006). The final expected volume (proportional to the invested biomass) of each generated structure was fixed to 0.3 m^3. The radius of the root system was fixed to 3 m around the tree, and was assumed to be large enough to consider any root contributing

Table 1. Four different types of root systems were simulated using the SIMUL3R program: heart-, tap- herringbone- and plate-like root systems

	Types of root system			
	Heart	Tap	Herringbone	Plate
Initiation				
Number of main lateral roots	7	5	0	5
Tap root	Yes	Yes	Yes	No
*f_growth*I*( )*				
Trajectory	Direction is given at insertion			
f_ram( )				
Distance between branching ΔL_i (m)	0.2	0.3	0.3	0.3
Horizontal/vertical	Uniform	40/60	60/40	40/60
Distance of the fork from origin ΔL_{fork} (m)	1.0	–	–	–
Branching on tap root	No	No	Yes	No
f_taper				
Taper equation	$\left(d_n^m - d_{n+1}^m\right)/dl = 12 \times \left(l_n^m\right)^{-0.6}$			
Taper of the tap root	Yes	Yes	No	–
f_ratio				
Mother/daughter relationship	$d_0^d = 0.45 \times d_n^m$			
Boundaries				
Rooting depth (m)	1.0	1.5	1.0	0.5
Radial distance (m)	3.0	3.0	3.0	3.0
Branching order	<4	<4	<4	<4

Morphological differences were generated by changing the nature of the morphological function. The number of the main apices initiated at the center of the bole was varied in the *INITIALISATION* procedure. Root systems could possess different branching patterns by altering the distances between branches, the direction of roots at the insertion point, or the forking of roots by using the *f_ram* function. Root taper could also have different values in *f_taper*, which emphasized the differences between rooting types. Finally, various types of boundary conditions highlighted differences between the four root morphologies. Data were taken from measurements carried out on root systems of different tree species explained in Dupuy (2003).

mechanically to tree stability i.e. roots situated at more than 3 m from the tree are assumed to have minor impact on uprooting.

In the *INITIALISATION* module, the number of main apices initiated at the centre of the bole was varied. Heart-like root systems possessed a maximum of seven main lateral roots whereas plate- and tap-root systems had only five main lateral roots. The herringbone-like structure did not have any main structural lateral roots attached to the bole but showed uniformly distributed finer lateral roots.

Branching characteristics were varied using different *model_ram( )* functions. Branching density was higher for the heart root system as the distance between branches was set to 0.2 m, whereas for all the other root systems this distance was set to 0.3 m. Similarly, the growth direction of roots at the insertion point was different in the tap- and plate-root systems, where 40% of roots were horizontal and 60% vertical. In herringbone patterns, 60% of roots were oriented horizontally and 40% vertically, whereas heart root systems possessed a continuous distribution of roots between the two directions. Forking was also present in heart systems, whereas lateral branching was present all along the tap root in the herringbone root systems.

The taper equation in *f_taper* was similar for every root system and was derived from real data from Maritime pine root systems (Danjon et al., 1999; Fourcaud et al., 2003a, b). The only exception concerned the main tap root of the herringbone system, which had no taper, typical of that found in adult poplar trees which have originated from cuttings (Dupuy, 2003). The relationship found between the *f_ratio* of the mother d_n^m and daughter d_0^d root diameter at the insertion point was linear and also derived from Maritime pine root system data.

Finally, the set of various types of boundary conditions enabled the differences between the four root morphologies to be appreciated. The

tap root system had a maximum rooting depth of 1.5 m, heart and herringbone root systems had a rooting depth of 1.0 m whereas the plate-like system was only 0.5-m deep.

Mechanical modelling

Numerical simulations of uprooting were carried out using the finite element software ABAQUS (Hibbitt, Karlsson & Sorensen, Inc., http://www.abaqus.com) and by following the methodology determined by Fourcaud et al. (2003b). The model was composed of two parts; the first being the root system and the second the surrounding soil.

Definition of the domain and meshing for FEM calculation

The root systems were described using a Multi-Tree Graph (MTG) format according to the topological rules given by Godin and Caraglio (1998). These root systems consisted of a set of slender axes connected together at specific branching points. Each axis was split into segments for which basal diameters, as well as spatial co-ordinates of their extremities were given. The finite element discretization with three-dimensional linear beam elements (Fourcaud and Lac, 2003) was performed automatically from the MTG file (Dupuy, 2003). The maximum size of beam elements was defined initially by the user.

The root systems were placed in a parallelepipedic block of soil, 7.0×7.0 m wide and 2.5-m deep. The surrounding soil domain was composed of one free top surface, four lateral faces for which horizontal displacements where imposed as zero and one bottom face that was blocked in all directions. The whole soil domain was discretized using 20-node quadratic brick elements.

Physical links between the two parts were taken into account considering stiff connections between the beam nodes and the closest brick nodes (Dupuy, 2003), i.e. no slipping or opening was allowed at the interface of the two domains.

Definition of materials

As this study was purely theoretical and parametric, the soil mechanical behaviour chosen was that of a simple, elastic, perfectly plastic model, i.e. without considering hardening. The yield criterion defining the limit of soil failure was given by the classical Mohr–Coulomb model $\tau = c - \tau \tan(\Phi)$, assuming that failure was controlled by the maximum shear stress τ and that this failure shear stress depended on the normal stress σ which is conventionally negative in compression. In this equation, c was the soil cohesion and Φ was the material angle of friction. The Mohr–Coulomb plasticity model provided in ABAQUS (see ABAQUS User's manual for details) was written in terms of three stress invariants in order to apply it to a general state of stress. Mohr–Coulomb yield criterion was classically defined by a straight line in the meridional plane and an irregular hexagonal section in the deviatoric plane. Nevertheless, the non associated flow potential was not a classical hexagonal pyramid, but was defined by a hyperbola in the meridional plane and a smooth deviatoric section (Menétrey and Willam, 1995). The elastic part of the soil constitutive law was assumed to be linear isotropic and governed by the Young's modulus E_{soil} and Poisson's ratio v_{soil}.

The anchorage of tree root systems was analysed in four different types of idealistic soils varying the two parameters c and Φ of the Mohr–Coulomb model (Table 2). These values represent a wide range of soil mechanical behaviours. Ideal cohesive soil, i.e. undrained clay soil, was represented with a null friction angle, a high cohesion value and a Poisson's ratio of 0.49 (Yang and Jeremic, 2002). Cohesion varied from 20.0 to 50.0 kPa and represent variations in clay shear resistance due to different moisture content or geomorphological origins. Young's Modulus (20.0 MPa) remained constant for all soil types. Ideal frictional soils i.e. drained sandy-like soils, were represented using a friction angle of 30°, a low cohesion value of 2.0 kPa and a Poisson's ratio of 0.3. The distinction between dry and saturated sands, according to effective stress analysis was studied by using a unit weight of 20.0 kN m^{-3} and 10.0 kN m^{-3} for dry and wet soil respectively, (Whitlow, 1995).

Root wood was considered as an elastic, brittle, perfectly plastic material considering the Mises yield surface with associated flow rule. Deformation prior to yielding was assumed to be linear elastic governed by the Young's Modulus $E_{\text{roots}} = 2.0$ GPa and Poisson's ratio $v_{\text{roots}} = 0.3$. The yield stress $\sigma_{\max}$ was taken to be 15.0 MPa.

Table 2. Four different types of idealistic soils were used for the analysis of anchorage efficiency: ideal frictional soil i.e. dry and saturated sandy sand and ideal cohesive soil i.e. and undrained clay-like soil behaviour

	Types of soil			
	Soft clay	Hard clay	Saturated sand	Dry sand
Young's Modulus (MPa)	20.0	20.0	20.0	20.0
Poisson's ratio	0.49	0.49	0.30	0.30
Cohesion (kPa)	20.0	50.0	2.0	2.0
Friction angle (°)	0.0	0.0	30.0	30.0
Volumetric weight (kN m^{-3})	20.0	20.0	10.0	20.0

The properties of each of these soil types were modelled using Mohr–Coulomb's elasto-plastic model.

Mechanical analysis

An infinitely rigid vertical lever arm was added to the top of the root system bole. This element did not aim to model the tree stem, but was only used in order to impose a rotation of the whole system. An initial calculation step was performed before simulating tree uprooting. This initial step allowed the initial geo-static stresses due to the self-weight of the whole system to be estimated. This phase is particularly important as the initial soil pressure has an impact on the soil shear strength according to the above Mohr–Coulomb criteria. The soil pressure increases linearly with depth; the deeper soil layers being more resistant to shear.

During the following step of analysis, a lateral displacement of 1.5 m was applied incrementally to a point of the upper rigid lever arm situated arbitrarily at a height of 10.0 m in order to apply a rotation to the root structure. The FEM analysis consisted of determining the mechanical response of the whole soil/root system, namely the reaction force calculated at the node where the displacement was imposed, the displacement of the discretization nodes, the elastic and plastic strains, as well as the stress field in both roots and soil.

Criteria for anchorage analysis

During the stem movement, the increments of reaction forces were recorded at the node where the displacement was imposed. The mechanical response of anchorage was then determined by drawing the force/displacement curve. The maximum force at the end of the lateral displacement was used to compare quantitatively the soil/root anchorage efficiency. Various variables such as displacement or plastic strain fields were monitored in the vertical, symmetrical plane defined by the direction of pulling and which seemed a logical choice for this study. This information was discussed in order to give a qualitative explanation of the uprooting mechanism.

Results

Characteristics of computed root systems

The heart-like root system was typical of that found in Red oak (Lyford, 1980). This structure was densely branched and the secondary lateral roots were oriented randomly between the horizontal and vertical directions. Seven main laterals were axi-symmetrically distributed around the vertical stem axis. The basal diameter of these lateral roots was 0.23 m. The distance between lateral branches ΔL was set to 0.20 m. Forks were generated at a distance of 1.0 m from the stem. The sum of the length of all roots (Σ length) was 160.0 m. The maximum rooting depth was 1.0 m and the root system horizontal diameter was 3.0 m.

The tap-like root system (Figure 3b) was typical of several coniferous species e.g. Maritime pine (Danjon et al., 1999) and Scots pine (*Pinus sylvestris* L., Köstler, 1968). The main roots were oriented either vertically or horizontally and did not develop forks. Five main laterals were connected to the stump with an axi-symmetric distribution. Their basal diameter was 0.20 m. The distance between two lateral branches on a mother root ΔL was set to 0.30 m (Coutts, 1987). The tap root basal diameter was 0.20 m. The total root length was 1.13 m. The depth of the root system reached a maximum of

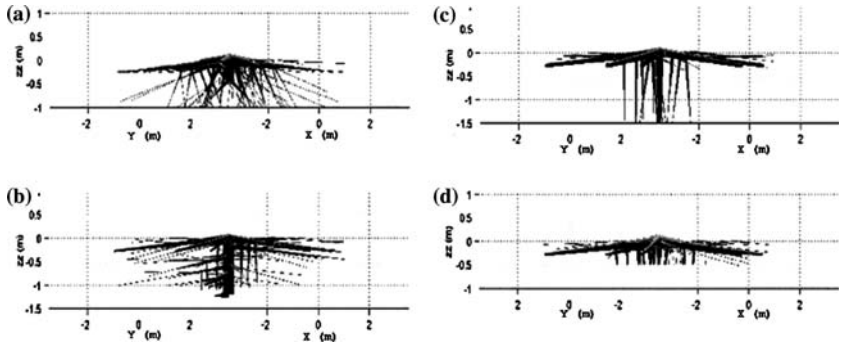


Figure 3. Four types of root systems denoting highly opposed rooting strategies were selected for the simulations: (a) the heart root system was 1.0-m deep and had forks; (b) the tap-root system was 1.5-m deep and constituted a thick first order vertical root and five main lateral roots; (c) the herringbone root system had a main vertical root of constant diameter, uniformly covered by second order lateral roots; (d) the plate like root system had no tap root and was 0.5-m deep.

1.5 m which corresponded to the length of the tap root. The radius of the whole structure was 3.0 m.

The herringbone-like root system (Figure 3c) possessed layers of horizontal second order lateral roots that were uniformly distributed along the main tap root. The distance between the whorls ΔL was 0.30 m. Each stage was composed of five laterals that were separated from each other by a constant angle. The total length was 170.0 m and the diameter of the main tap root was 0.17 m. The total depth of the root system was equal to 1.0 m and the total horizontal diameter was 3.0 m.

The plate-like root system (Figure 3d) was representative of that found in Sitka spruce (Köstler et al., 1968) or Norway spruce (Puhe, 2003; Stokes et al., 2006). This system did not have a tap root, but five main laterals only, which were attached to the stump. The total length of the root system was 80.0 m. The second order laterals had a basal diameter of 0.27 m and bore sinkers, separated by a constant distance ΔL equal to 0.30 m. The total depth of the root system was limited to 0.50 m, whereas the total diameter was 3.0 m.

Qualitative results

During the uprooting simulations, forces were transmitted by the roots into the soil inducing deformation of the entire root/soil plate. The roots that were located on the leeward side, i.e. the side of the tree held in compression during overturning, tended to plunge downwards, which induced a local increase in soil pressure (Figure 4). Simultaneously, windward roots on the opposite side of the tree moved upwards out of the soil. This movement resulted in a decrease in soil pressure on this side of the root system. After a critical displacement of the stem, the failure yield criterion was reached in certain regions of the soil. These failure areas were highlighted by the field of equivalent plastic strain (Figure 5) and took the form of a slip surface.

Very different uprooting mechanisms were observed in sand and clay soils (Figures 4 and 5). In clay soil, the root/soil system rotated around an axis that was situated straight below the stem. Upward displacements of the windward side were almost equal to downward displacements on the leeward side. The slip surface was symmetric and roughly circular. The diameter of the slip surface was lower in the hard clay than in the soft clay and root plastic strains occurred on large diameter lateral roots close to the bole centre.

In sandy soil, the soil/root system rotated around an axis that was shifted leeward. Permanent strains expanded further away on the windward side, compared to the leeward side. The slip surface was significantly larger than in clay soils.

Plastic strain profiles in the soil were also very different according to the type of root systems considered (Figure 5). Heart type systems showed a rather moderate and homogeneous distribution of permanent strains around the centre of the bole. Tap root systems showed localised

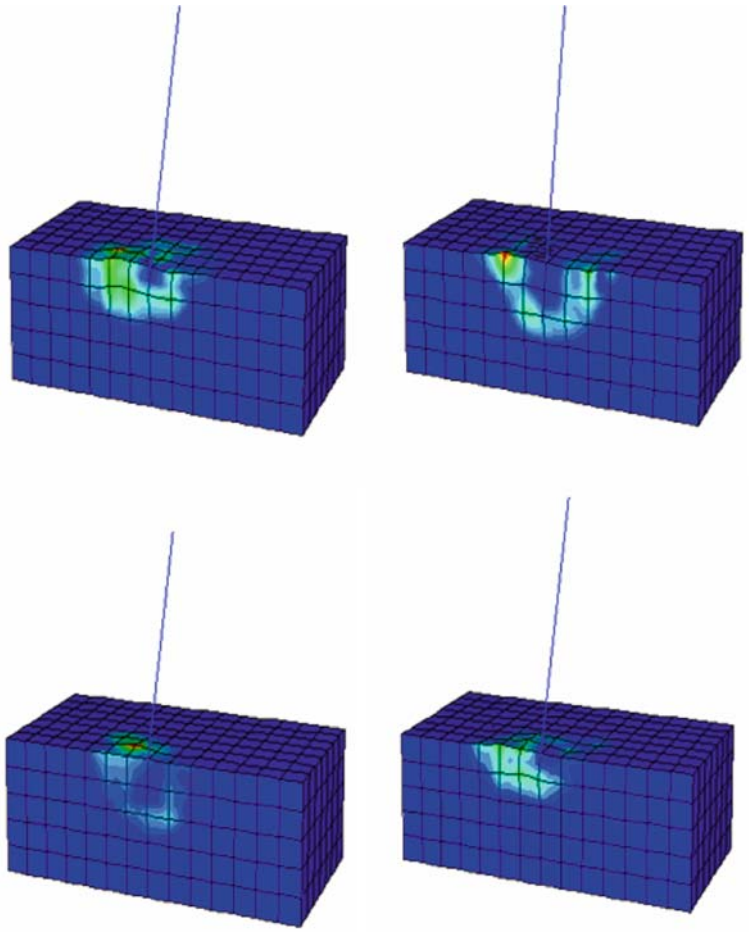


Figure 4. Fields of plastic strain in clay soil (ideal cohesive material with low high resistance): (a) heart-, (b) tap-, (c) herringbone- and (d) plate-like root systems.

slip surfaces extending deeply in the soil. The herringbone root system mobilized only a slight amount of soil during uprooting, indicating that most failure occurred in the roots close to the main vertical root. Soil failure in plate-like root systems was relatively shallow.

Quantitative results

The response curves resulting from the uprooting FEM simulations were typical of plastic materials (Figure 6). The maximum resistive moment calculated at the bole for a range of imposed displacements for all root patterns embedded in both soil types were characteristic of root anchorage (Figure 7).

Resistance to displacement was higher in clay soils (221 ± 37 kN m) compared to sandy soils (127 ± 13 kN m) (Figure 7, means are ± standard deviation). This result was particularly evident in heart (315 kN m in clay and 129 kN m in sand) and tap root systems (292 kN m in clay and 175 kN m in sand). However, differences in the overturning resistance of herringbone and plate root systems were low when clay and sandy soils were compared (Figure 7). The decrease of shear resistance in clay soils induced a mean decrease of 32% of the global resistance to uprooting. However, water saturation in the sandy soil reduced root anchorage strength by 15%.

Whatever the soil characteristics, heart (222 kN m) and tap (234 kN m) root patterns were the most resistant when compared to her-

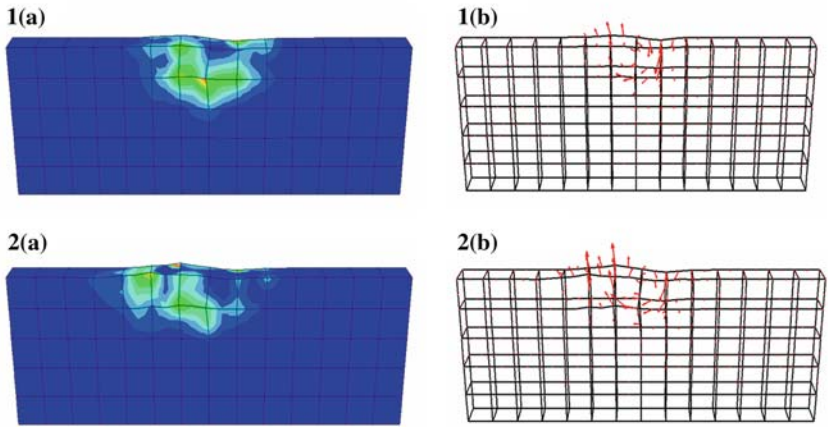


Figure 5. (a) Plastic strain and (b) displacement field observed in a strip of (1) hard clay material (2) dry sand for the plate root system.

ringbone (122 kN m) and plate (117 kN m) root systems. However, the tap-root system was best anchored in sandy soils (152 kN m in saturated sand and 198 kN m in dry sand) whereas the heart root system was more efficient in soft clay (265 kN m) (Figure 7). The weakest anchorage was found to be that of the plate root system, even though this architectural pattern was slightly more resistant than the herringbone root system embedded in hard clay.

Discussion

The model developed was relatively simple, easy to use and did not require long calculation times to run each simulation. Results for bending moment were realistic and comparable to those found in experimental studies (Cucchi et al., 2004; Stokes et al., 2005). As a tool for studying root anchorage, the model was useful in that branching could be determined by the user, by setting certain architectural parameters. The influence of any particular parameter can then be estimated, as volume, material and soil properties remain constant. The model could be adapted for testing herbaceous species e.g. maize (*Zea mays* L.) which possess a nodal type root system, or even rhizome or bulb species. Similarly, the influence of many different soil types on uprooting could also be tested.

Influence of soil type

The soil type had a strong influence on the resistance to uprooting as well as on the mechanisms occurring during failure. In clay soil, the shear resistance was constant anywhere in the soil and was independent of the isostatic stress due to the chosen null friction angle. Therefore, areas of plastic strain were comparable on both windward and leeward sides of the tree, which thus resulted in a symmetric mechanism (Figure 4a). In sandy soils however, the shear resistance was severely reduced with the decrease in pressure in the soil. Consequently, the field of plastic strain on the windward side of the tree was much more developed, reaching the distal regions of the soil (Figures 4b, 5). On the leeward side of the tree however, the plastic yield remained closer to the stump (Figure 5). The dissymmetry of this type of uprooting mechanism has been highlighted by experimental studies on tap-rooted Maritime pine growing in sandy soils (Cucchi et al., 2004; Stokes, 1999) and also in *Pinus peuce* (Mickovski and Ennos, 2003) even though it has never been related to soil internal friction. For most of the situations found in nature however, the soil is neither pure clay nor pure sand and the presence of such dissymmetry, with a larger amount of windward roots being uprooted, will exist. This type of anchorage failure has been already been described by Mickovski and Ennos (2002), who

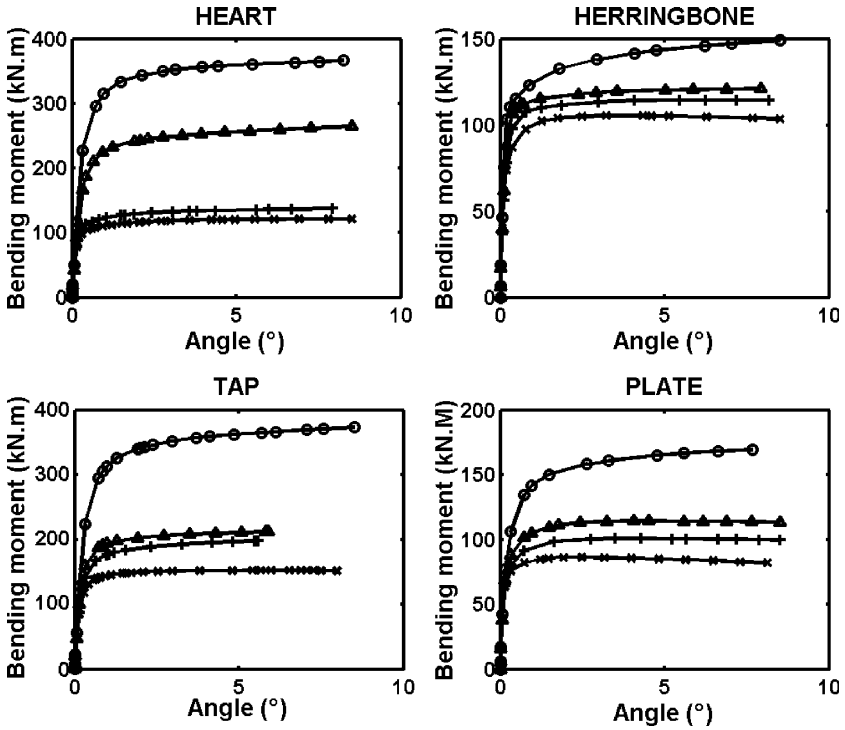


Figure 6. Bending moment vs. rotation angle during the simulation of the four root systems, heart, herringbone, tap and plate, in four different soil types, 'dry sand' (ideal frictional soil in dry conditions) (+), 'saturated sand' (saturated drained behaviour of frictional soils) (x), 'soft clay' (ideal cohesive material with low shear resistance) (Δ) and 'hard clay' (○) (ideal cohesive material with low high resistance). Moment/rotation curves are typical of plastic behaviour.

carried out winching tests on Scots pine growing in a clay loam soil. These authors observed a complex network of cracks in the soil on the windward side of the tree during overturning.

Previous investigations carried out by Crook et al. (1997) showed that uprooting dissymmetry is not only influenced by the soil type but also by rooting depth. This observation is confirmed when observing the qualitative results that were obtained on the tap-root system (Figure 5). This root system was deeper than the others tested and did not exhibit any strong dissymmetry in sandy soils when compared to e.g. the plate root system.

The effect of soil shear resistance tended to decrease the radius of the slip surface and reduce the area where soil is resisting failure through shearing (Mattheck and Breloer, 1994; Yang and Jeremic, 2002). Breaking of roots occurs near the centre of the bole where roots were thicker. The total amount of roots resisting the bending moment was thus increased during uprooting. A similar phenomenon was also observed by Stokes et al. (2000), who carried out winching tests on beech (*Fagus sylvatica* L.), with heart-like root systems growing on a clay loam.

Influence of root morphology

The mode of failure derived from the numerical simulations was strongly determined by root morphology as has also been observed in experimental studies (Coutts, 1986; Crook et al., 1997; Stokes et al., 2000; 2005a). Heart like root systems generally had the most efficient anchorage, which is in agreement with results from experimental investigations (Stokes et al 2000, 2005a, b). These root systems possess large lateral roots originating from the centre of the bole, which then rapidly branch into smaller roots. The heart-like architecture therefore combines stiffness close to the trunk and dense fibrous networks further away, which improve soil shear resistance (Wu et al., 1988). When forked roots are lifted up out of the soil, they also carry soil upwards with them in the crux of the fork, the

weight of which will help increase overturning resistance (Stokes et al., 1996).

The tap root system is a particularly efficient structure on sand due to the increased rooting depth. Depth is probably the most important factor on sandy soils because shear resistance increases with the earth's pressure, due to changes in friction angle (Whitlow, 1995). On the contrary, the herringbone structure was not very resistant to uprooting due to a lack of stiff secondary laterals. Niklas et al. (2002) illustrate the lack of efficiency of a massive tap root if not associated to thick lateral roots, as soil is not mobilized around the root system. The insertion angle of lateral roots on the tap root is also important, with an optimal angle around 60° (Stokes et al., 1996). The influence of this branching angle was also not examined in this study, but the model developed would be a useful tool to investigate these parameters further.

The least resistant rooting type was the plate like root system. Despite possessing massive lateral roots, only a small amount of soil was mobilized during uprooting due to the limited penetration of vertical roots. If there are not many vertical roots embedded in the soil, the soil/root cohesion will be decreased and anchorage strength reduced. Therefore, when plate root systems overturn, the entire soil/root plate is lifted upwards on the windward side (Mattheck and Breloer, 1994). The inefficiency of plate root systems to overturning has also been demonstrated several times through static winching tests (Nicoll et al., 1995; Peltola et al., 2000; Stokes et al., 2000, 2005a, b).

Our studies of root anchorage were carried out on root systems which had the same diameter (6.0 m), but different depths (0.5 m, 1.0 m and 1.5 m). As a consequence, the influence of depth could be easily transposed to the analysis of a shape effect, i.e. the influence of the plate radius/plate depth ratio on root anchorage. However, one can expect a significant scale effect since the root material biomass was fixed to a constant value. This scale effect has not been investigated here, but must be considered in future analyses. Many aspects of the uprooting mechanism still need studying, in particular, the influence of root system volume and diameter. In our study, these parameters were fixed to

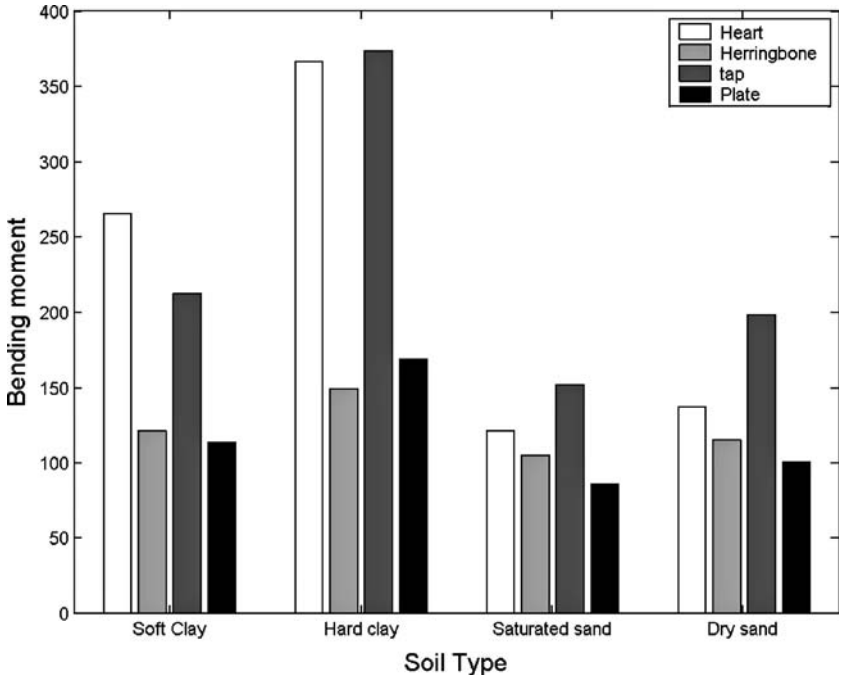


Figure 7. Maximum bending moment of four different root systems in four theoretical soils. The deep rooted tap system was the most resistant to uprooting in sandy-like soil, whereas the heart root system was the most resistant in clay-like soils.

dimensions assumed large enough not to have an effect on uprooting. Nevertheless, boundary conditions, whether horizontal or vertical, should be taken into consideration. In nature, the boundary conditions which influence uprooting include obstacles in the soil e.g. stones or bedrock, changes in soil density and waterlogging, resulting in the death of root tips (Cucchi et al, 2004). These parameters translate into complex mechanical and physical phenomena, which influence root/soil frictional resistance, soil plastic flow around the root–soil plate and temporal changes in soil properties. Modelling the influence of such boundary conditions would provide highly useful information in an area where these data are scanty.

The results provided by this study are new qualitative elements for the understanding of uprooting failure in real trees. Phenomena such as stump breakage, root/soil plate asymmetry or overturning of the soil/root plate were explained by physical and geometrical factors. It should be pointed out however, that the root/soil configurations used in our study were theoretical and cannot be found in nature as such. Particularly, root growth is governed by soil type and identical species can produce different root morphologies and volumes depending on local soil conditions (Fitter, 1987). Soil behaviour was also idealistic and uniformly distributed, whereas real soils exhibit strong heterogeneities. Nevertheless, our investigation demonstrated that root architecture and soil type are major factors governing root anchorage, even when volume and material properties are constant between each rooting type.

Although certain simplifications were made in this study, the model developed proved useful for studying different characteristics of root anchorage with regards to root architecture. In the future, certain parameters could be modified to determine the effect of e.g. root wood strength (Stokes and Mattheck 1996), root shape (Nicoll and Ray, 1996), root asymmetry (Coutts et al., 1999), root slippage through soil, soil type (Moore, 2000), soil hydrology, on tree root anchorage. Not only could these parameters be modified, but the model could also be used to test the effect of cyclic behaviour e.g. wind gusting on tree anchorage. Further studies should include the comparison of field data with results from the model. However, as it is impossible to simulate field conditions exactly, highly controlled experiments should be set up (Hamza et al., 2006). A considerable improvement in both the hypotheses used for modelling and the understanding of uprooting mechanisms could therefore be obtained.

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